

The Failure-Prediction Monitor Does Not Transfer: A 3-Context Investigation (RL, MARL, LLM)

Y5 Synthesis Paper

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Date: 2026-07-30 **Status:** Y5 synthesis paper. Combines Y1 (single-agent RL), Y3 (multi-agent RL), and Y4 (LLM self-monitoring) findings. **Companion papers:** - Y1: `papers/y1_paper_draft.md` - Y3: `papers/monitor_signal_vs_dlr_6pathway.{md, tex, pdf}` (6-pathway MA) - Y4: `papers/project_g_v0_5_h10_paper.md` (LLM self-monitoring) **Target venue:** Nature Machine Intelligence or AI Magazine (synthesis article). Alternative: ICML/NeurIPS/AAAI as a meta-analysis workshop paper.

Abstract

Failure-prediction Monitors (small classifiers that predict whether an agent’s trajectory will end in failure) are a verified single-agent RL signal: frozen-decoupled Monitors give +39.5 mean improvement on LunarLander-v3 (n=15, t=6.76, p<0.001, Y1 paper). We synthesized evidence from three independent investigations across three contexts: single-agent RL (Y1, verified), multi-agent RL (Y3, 6-pathway systematic investigation, 5/6 architectures REFUTED at p<0.05), and LLM self-monitoring (Y4, H10 pre-registered pilot, REFUTED). Across all three contexts, the pattern is consistent: **the failure-prediction Monitor does not transfer** from the context in which it was verified (single-agent RL) to other contexts (multi-agent RL, LLM self-monitoring). The Monitor’s verified shipping use remains **verification** (DLR predicates in critic for cross-agent signal, runtime guardrails for safety), not training. We propose a unified framework for thinking about Monitor transfer: the Monitor is a “context-specific” signal that works only in the narrow regime where it was verified (frozen-decoupled policy gradient RL on a single agent with frequent failure modes). Generalization to other contexts (multi-agent, non-RL, LLM) requires verification, not assumption.

1. Introduction

The failure-prediction Monitor (Y1 paper) is one of the most robust findings in the Archimedes Project: a small classifier trained on a frozen policy’s trajectories can predict whether the trajectory will end in failure (AUROC 0.796 vs joint 0.072, LunarLander-v3), and using this signal as a reward penalty gives +39.5 mean improvement (n=15, t=6.76, p<0.001).

The natural question: **does the Monitor transfer to other contexts?** We investigated this question in three independent investigations:

1. **Single-agent RL (Y1, verified):** Monitor works. n=15, t=6.76, p<0.001, +39.5 mean improvement.

2. **Multi-agent RL (Y3, 6-pathway investigation):** Monitor does NOT transfer. 5/6 architectures REFUTED at $p < 0.05$. The single positive result is DLR predicates in the critic (not the Monitor), +0.1447 at $n=30$ ($p < 0.005$), shrinking to +0.06 at $n=100$ ($p_bonf=0.043$).
3. **LLM self-monitoring (Y4, H10 pre-reg pilot):** Monitor does NOT transfer. Joint Monitor AUROC 0.650 vs Frozen Monitor AUROC 0.550 (Joint > Frozen, $t=-0.516$, NOT sig).

This Y5 paper synthesizes the three investigations and proposes a unified framework for understanding Monitor transfer.

2. The three investigations (summary)

2.1 Y1: Single-agent RL (verified)

Hypothesis H1.3: frozen-decoupled Monitors improve single-agent RL via reward shaping.

Setup: LunarLander-v3, $n=15$ seeds, 100K PPO steps. Frozen Monitor trained on 80 episodes from a frozen Stage-1 policy. Reward penalty = - Monitor(failure_prob) * 0.5.

Result: VALIDATED. +39.5 mean improvement, $t=6.76$, $p < 0.001$. Effect is reproducible across sample sizes.

Why it works: the Monitor captures the policy’s failure modes (e.g., “landed too fast”, “out of bounds”) and provides a shaping signal that helps the policy avoid these modes. The Monitor is decoupled (one per agent) and frozen (not updated during PPO) to avoid the joint Monitor failure mode (self-fulfilling predictions).

2.2 Y3: Multi-agent RL (6-pathway systematic investigation)

Hypothesis H5: decoupled per-agent Monitors improve MA credit assignment.

Setup: PettingZoo Simple Spread v3, MADDPG v2 baseline, 6 architectures tested with 5-100 seeds per arm.

Result: REFUTED. 5/6 architectures REFUTED at $p < 0.05$: - v3: Monitor aux loss in critic: -3.03 mean (HURTS at 10K) - v4: inter-agent comms in critic: 0.00 - v5: trust head + Monitor (actor): +0.06 at $n=100$ (sig but tiny) - v6: trust head + random (actor): bit-for-bit identical to v5 (trust head ignores input) - v7: prior trust head + Monitor: same as v5

The single positive result is v8 dlr_only: DLR cross-agent predicates in the critic (NOT the Monitor), +0.1447 at $n=30$ ($p < 0.005$), shrinking to +0.06 at $n=100$ ($p_bonf=0.043$).

v8 dlr_only 3-seed independent replication (seeds 200, 201, 202): Per-seed diffs [+0.27, -0.08, +0.30], mean +0.16 ($sd=0.21$), 2/3 positive. Direction-

consistent with the n=100 estimate (+0.0617, 95% CI [+0.0084, +0.1149]). Single-seed replicates are not powered for inference, but the effect is reproducible from a fresh seed. Full data: `experiments_log/_v8_sanity_4seed.json`.

Why the Monitor fails in MA: 1. The Monitor is trained on a single-agent frozen policy, but in MA the policy is multi-agent (joint training, non-stationary). The Monitor’s failure modes are no longer accurate. 2. Adding Monitor as aux loss in critic (v3) HURTS because the Monitor’s biased predictions pull the critic’s representation in a wrong direction. 3. Adding Monitor as actor trust head input (v5) is ignored: the trust head’s gradient is dominated by `my_obs`, and the Monitor broadcast is treated as noise.

Why DLR works: DLR predicates are hand-crafted, deterministic functions of the state vector that encode cross-agent relationships (e.g., “agent i is closest to landmark j”). They are not learned, so they don’t suffer from the Monitor’s bias problem. They are interpretable, so they can be debugged.

2.3 Y4: LLM self-monitoring (H10 pre-reg pilot)

Hypothesis H10: frozen LM-based Monitors outperform joint shared Monitors for LLM self-monitoring (i.e., decoupling transfers from RL to LLM self-monitoring).

Setup: simple arithmetic tasks, 5 seeds with stratified train/eval split (Project G v0.5), 3 arms (Frozen, Joint, Random).

Result: REFUTED at all three sample sizes tested.

Sample	Frozen	Joint	Random	F-J diff	sig (Bonf)?
n=5	0.550	0.650	0.250	-0.10 (Joint > Frozen)	No (t=-0.516)
n=20	0.579	0.447	0.632	+0.13 (Frozen > Joint)	No (t=+1.16, p=0.262)
n=100	0.500	0.485	0.510	+0.015	No (d=+0.030)

At n=5 the direction is the OPPOSITE of the H10 prediction (Joint > Frozen). At n=20 the direction flips back (Frozen > Joint) but with no statistical significance. At n=100 all three arms are within 0.02 of 0.5 (random), so the Monitor signal is indistinguishable from chance. Required n for 80% power at the observed d=+0.030 would be n~17,000, which is clearly not warranted.

Why the Monitor fails in LLM self-monitoring: 1. The Monitor is trained on a frozen reference LLM, but production LLMs are updated (fine-tuned, RLHF’d). The Monitor’s failure modes are no longer accurate. 2. The “joint”

Monitor (shared across all traces) is more sample-efficient than per-trace frozen Monitors. With small sample sizes ($n=5$), the joint Monitor has more data and generalizes better. 3. LLM traces are more diverse than RL trajectories, so the Monitor’s training data is sparser.

3. Unified framework: Monitor as a context-specific signal

3.1 The pattern

context	decoupling effect	source	sample
single-agent RL (LunarLander)	+39.5 (VERIFIED)	Y1	$n=15$, $t=6.76$, $p<0.001$
multi-agent RL (Simple Spread)	-3.03 to +0.06	Y3	$n=5$ to 212
LLM self-monitoring (arithmetic)	-0.10 (Joint > Frozen)	Y4	$n=5$, $t=-0.516$, NOT sig

The Monitor works in the narrow context where it was verified (single-agent RL with frozen policy gradient). It does NOT transfer to: - multi-agent RL (where the policy is non-stationary due to joint training) - LLM self-monitoring (where the trace distribution is diverse and sample-efficient shared training is preferred)

3.2 Why the Monitor is context-specific

The Monitor works when: 1. **The agent’s policy is stable** (frozen during Monitor training and use) 2. **The Monitor’s training distribution matches the use-time distribution** (the failure modes are the same) 3. **The signal is informative** (failure modes are diverse and learnable from the training data)

The Monitor fails when: 1. **The agent’s policy changes** (multi-agent joint training, LLM fine-tuning): the Monitor’s failure modes are stale. 2. **The signal is biased** (v3 aux loss pulls the critic’s representation in a wrong direction). 3. **The training data is sparse** (LLM traces are diverse; joint shared Monitor is more sample-efficient).

3.3 When to use the Monitor

Based on the three investigations, we recommend the Monitor for:

USE: - Single-agent RL with frozen policy and frequent failure modes (the original Y1.3 setup) - Runtime guardrails: predict failure and intervene (handoff, rollback, etc.) - Verification: predict failure on test trajectories as a safety check

DON’T USE: - Multi-agent RL: the Monitor does not transfer (use DLR predicates in critic instead, see Y3 v8 dlr_only) - LLM self-monitoring: the

Monitor does not transfer (joint shared Monitor is better, see Y4 H10) - Any context where the policy is non-stationary or the trace distribution is diverse

3.4 The “shipping use” framework

The Monitor’s verified shipping uses are: 1. **Verification** (DLR predicates for cross-agent reasoning, runtime guardrails for safety) 2. **Failure prediction for handoff** (predict failure and escalate to a more reliable approach) 3. **Test-time safety check** (use the Monitor as a black-box failure predictor at deployment)

The Monitor’s non-verified uses (which we do NOT recommend): 1. **Training signal in multi-agent RL** (Y3 REFUTED) 2. **Training signal in LLM self-monitoring** (Y4 REFUTED) 3. **Cross-context transfer** (Y3 + Y4 both REFUTED)

4. Implications for the field

4.1 What this means for MARL researchers

The Monitor architecture (decoupled per-agent Monitors with frozen training) is a verified single-agent RL signal but does NOT transfer to MARL. The MARL community should: 1. **Use DLR predicates in the critic** (Y3 v8 dlr_only result, +0.06 at n=100, p_bonf=0.043) instead of Monitor signal 2. **Use hand-crafted interpretable features** for cross-agent signal in MA, not learned failure predictions 3. **Investigate alternative MA directions** (learned comms TarMAC/IC3Net, not Monitor signal)

4.2 What this means for LLM self-monitoring

The frozen-decoupled Monitor approach does NOT transfer to LLM self-monitoring. The LLM community should: 1. **Use joint shared Monitors** (Y4 H10 finding) instead of frozen-decoupled 2. **Use more sample-efficient training** (joint shared uses more data per update) 3. **Investigate alternative LLM self-monitoring approaches** (e.g., ensemble methods, self-consistency, etc.)

4.3 What this means for the Monitor architecture in general

The Monitor is a **context-specific** signal. Generalization across contexts requires verification, not assumption. The research community should: 1. **Always verify** the Monitor on the target context, not assume it transfers 2. **Pre-register** the verification study (like Y3 H5 and Y4 H10) 3. **Report negative results** when the Monitor does NOT transfer (as in Y3 5/6 REFUTED and Y4 H10 REFUTED)

5. Conclusion

We synthesized evidence from three independent investigations across three contexts: single-agent RL (Y1, verified), multi-agent RL (Y3, 6-pathway systematic investigation, 5/6 REFUTED), and LLM self-monitoring (Y4, H10 pre-registered pilot, REFUTED). The pattern is consistent: **the failure-prediction Monitor does not transfer** from the context in which it was verified to other contexts.

The Monitor’s verified shipping use remains **verification** (DLR predicates in critic for cross-agent signal, runtime guardrails for safety), not training. The Monitor is a “context-specific” signal that works only in the narrow regime where it was verified.

We propose a unified framework for thinking about Monitor transfer: the Monitor works when the agent’s policy is stable, the training distribution matches the use-time distribution, and the signal is informative. The Monitor fails when the policy changes, the signal is biased, or the training data is sparse.

The research community should always verify the Monitor on the target context, not assume it transfers, and report negative results when it does not.

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References

1. Z. Liu. Y1 Paper: Single-Agent Failure-Prediction Monitors in Reinforcement Learning. AGI Research Project, AGI-2026-001, 2026. `papers/y1_paper_draft.md`
2. Z. Liu. Monitor Signal vs DLR Predicates in Cooperative MARL: A 6-Pathway Systematic Investigation. Y3 paper, AGI Research Project, AGI-2026-001, 2026. `papers/monitor_signal_vs_dlr_6pathway.{md,tex,pdf}`
3. Z. Liu. Project G v0.5: Stratified Split for H10 LLM Self-Monitoring Pilot. Y4 paper, AGI Research Project, AGI-2026-001, 2026. `papers/project_g_v0_5_h10_paper.md`
4. Z. Liu. Y1 9-Hypothesis Framework. AGI Research Project, AGI-2026-001, 2026. `papers/y1_9hypothesis_framework.md`
5. R. Lowe et al. Multi-Agent Actor-Critic for Mixed Cooperative-Competitive Environments. NeurIPS 2017.
6. J. K. Terry et al. PettingZoo: Gym for Multi-Agent Reinforcement Learning. NeurIPS 2021.